SARC Authority Derivation

Final Gap Plan: From 9.2 to 9.5

Repository assessed at HEAD **12414e14f2c71ade1f77a023aa1c38de195485cd**

Current assessment: **9.2/10** - approximately **9.3** once the latest Python 3.12 CI job completes green.

Bottom line	The repository already has enough scientific substance to reach 9.5. The remaining work is consolidation, product-grade packaging, one realistic cost-sensitive result, one larger non-planted domain, and independent reproduction. Do not add another major SARC concept before these are complete.

What changed since 7.9	Status
Realistic core != reduct domain	Done - code/cloud model has a 6-attribute insufficient core and two distinct 7-attribute reducts.
Isolated policy trajectories	Done - arm-local budget, grant ledger and admission history; null delta reported honestly.
SAT / MaxSAT / weighted synthesis	Done - exact on feasible models.
Scaling	Done materially - registered 30/60/100-property families; exhaustive search times out while SAT/MaxSAT solve quickly.
AuthorityBench	Done - 3 domains x 4 baselines x 6 metrics.
Hermetic CI dependencies	Done in workflow.
Mutation as release gate	Done - release-check invokes mutation.

Prepared as an execution document: each remaining gap has an implementation action, a test, and a stop condition.

1. The 9.5 acceptance standard

A 9.5 repository is not merely technically correct. It must be installable, independently reproducible, scientifically differentiated, operationally meaningful, and easy for an external researcher to interrogate.

Requirement	Current state	9.5 gate
Clean CI	Python 3.11 current HEAD green; Python 3.12 in progress at review time	Both latest-HEAD jobs green
Mutation gate	Hard-gated in release-check	Remain green after all remaining changes
Core/reduct theory	Corrected and machine-checked	No regression
Realistic core != reduct	Present in v4 code/cloud domain	Keep as central evidence
Symbolic synthesis	SAT / cardinality MaxSAT / weighted MaxSAT	Package and expose as stable API
Scaling	100 candidate attributes in planted hard families	Add one larger non-planted domain
Weighted governance context	Synthetic demonstration only	Realistic domain where cost changes selected contract
AuthorityBench	3 domains, 4 baselines	Document and expose as canonical benchmark
Installability	Incomplete / ambiguous	Wheel install in fresh venv and API smoke test
External reproduction	Author/AI-assisted only	Independent run from clean clone or external PR/reproduction report

Score rule	Do not raise the score because wording improves. Raise it only when a material uncertainty disappears through new evidence.

2. P0 - Make the Authority Compiler a real installable package

This is the most immediate remaining engineering gap. The repository now has **src/authority_compiler/api.py**, but the packaging declaration exposes only the top-level **authority_bench** module. Pytest can import **authority_compiler** because **src/** is injected into the test path; that does not prove that a built wheel contains the package.

Required changes

- Configure setuptools package discovery for the src layout, e.g. package-dir = {"" = "src"} plus package discovery under src.
- Export the canonical API from authority_compiler.__init__: derive_authority_contract, AuthorityContract, and stable result types.
- Keep authority-bench as a console script, but do not make it the only installed artifact.
- Build both sdist and wheel in CI.
- Install the wheel into a fresh temporary virtual environment and run a true black-box smoke test.

Mandatory CI smoke test

```
python -m build python -m venv /tmp/ac-smoke /tmp/ac-smoke/bin/pip install dist/*.whl
/tmp/ac-smoke/bin/python - <<'PY' from authority_compiler import derive_authority_contract,
AuthorityContract print("IMPORT_OK") PY /tmp/ac-smoke/bin/authority-bench --help
```

Acceptance criterion	A contributor with only the built wheel must be able to import authority_compiler and run one canonical contract derivation without relying on the repository root, pytest pythonpath, editable installs, or sibling source directories.

Also add a package-level version and a minimal public API stability statement. At 9.5, the repository should behave like a research library, not only like a paper artifact.

3. P0 - Reposition the paper and README around the actual contribution

The repository has outgrown the current framing. The live paper still centers the historical v2 result: a core attribute set that happens to be sufficient and unique on one registered model. That is no longer the strongest result.

The central research object should now be:

New central claim	Given explicit loss semantics and executable reachability, synthesize and certify the least-cost sufficient runtime context required to govern an autonomous system.

Recommended paper architecture

1. **Problem:** hand-authored authority schemas are incomplete or over-inclusive.
2. **Formal object:** reachable loss-discriminating state pairs and discernibility sets.
3. **Core vs reduct:** individual indispensability is not joint sufficiency.
4. **Realistic evidence:** code/cloud domain with insufficient 6-attribute core and two alternative 7-attribute reducts.
5. **Synthesis:** SAT, MaxSAT and weighted authority contracts.
6. **Scale:** 30/60/100 candidate-attribute hard families where exhaustive search becomes infeasible.
7. **Benchmark:** AuthorityBench across procurement, code/cloud, and data/communications.
8. **Runtime implication:** minimum sufficient governance context can be compiled into gate schemas and certificates.
9. **Limitations:** declared loss-model completeness, reachability-model validity, partial observability, temporal/probabilistic losses, external validation.

README first-screen redesign

- One-line thesis.
- One realistic code/cloud example showing core != reduct.
- One command for authority-compiler.
- One command for authority-bench.
- Three headline results: realistic multiple reducts; symbolic scaling; cost-aware synthesis.
- Then historical v1/v2 correction story below the fold.

Stop condition	A new reader should understand in under 60 seconds that this is now a governance-context synthesis engine and benchmark - not primarily a paper about P*.

4. P1 - Produce one realistic cost-sensitive governance result

Weighted synthesis exists, but the current real domains do not yet demonstrate the result that would make 'Minimal Sufficient Governance Context' scientifically compelling. The synthetic XOR example proves the mechanism, not its operational value.

Design a realistic case where minimum cardinality and minimum operational cost disagree.

Recommended domain: code/cloud authority

Candidate observation	Illustrative cost components
branch	Local repository metadata; low latency; low privacy
environment	Remote deployment-control lookup; medium latency; stale-state risk
approval_token	Remote approval service; high availability requirement
resource_owner	IAM/CMDB lookup; medium latency; possible staleness
delegated_role	Local identity context; low latency
data_classification	Policy metadata lookup; medium latency/privacy

Use a preregistered multi-objective cost model, for example:

```
cost(p) = alpha * normalized_latency_ms(p) + beta * privacy_exposure(p) + gamma * staleness_risk(p) +
delta * lookup_failure_probability(p)
```

Then compare:

- minimum-cardinality reduct;
- minimum-cost reduct;
- all alternative reducts;
- resulting latency/privacy/staleness profile;
- safety equivalence certificate.

Required result shape	At least one realistic registered domain should produce two or more sufficient contracts where the minimum-cost contract differs from the minimum-cardinality contract, while both remain exactly sufficient over the registered reachable model.

Do not tune costs after seeing the solver output. Register the cost rationale and values before execution.

5. P1 - Add one larger non-planted realistic domain

The 100-property scaling result is useful, but the family is deliberately planted and hand-derived. AuthorityBench's realistic domains remain small at 8-10 candidate properties. A 9.5 research artifact should contain at least one larger realistic model where the reduct is not designed in advance.

Recommended target

- 30-50 candidate context properties.
- A realistic code/cloud or enterprise multi-agent action domain.
- Reachability generated from workflow constraints, tool capabilities, delegation state, approval state, environment topology and data sensitivity.
- Loss predicates authored before synthesis and frozen by preregistration.
- No hand-planted reduct.
- Solver discovers the minimum-cardinality and minimum-cost contracts.
- If the result is trivial or unique, keep it. Do not redesign until an interesting answer appears.

Suggested candidate-context families

Family	Examples
Identity/delegation	actor, role, tenant, delegation depth, issuer, audience, approval state
Resource	repository, service, environment, owner, criticality, region, data class
Action	operation, side-effect class, reversibility, write scope, blast radius
Workflow	branch, deployment stage, change window, incident state, release train
Evidence/state	policy version, ticket state, risk score, secret exposure, rollback availability

Acceptance criterion	The solver should operate on a non-planted, preregistered realistic domain of at least 30 candidate context fields and produce a certified sufficient contract without exhaustive power-set enumeration.

6. P1 - External reproduction

The repository is now heavily self-audited and AI-assisted. That is not a defect, but it becomes the remaining credibility ceiling. A genuine 9.5 should include evidence produced by someone who did not design the system.

Minimum acceptable external validation

- Fresh clone on a machine/environment not used during development.
- Run the documented bootstrap and canonical reproduction path.
- Verify package installation from built wheel.
- Run at least one AuthorityBench domain.
- Run the code/cloud multi-reduct example.
- Record exact commit SHA, OS, Python version, commands, elapsed time and output hashes.
- Submit findings as a GitHub issue, PR, signed reproduction note or independent report.

Stronger external validation

- An external contributor adds a fourth AuthorityBench domain.
- A formal-methods/security researcher challenges the novelty fence or theorem framing.
- A user applies authority_compiler to a domain not designed by the author and obtains a useful contract.

Score implication	9.5 can be justified with one clean independent reproduction plus the technical gaps above. 10/10 would require stronger external evidence such as adoption, citation, external contributions, or peer-review success.

7. Claude Code execution sequence

Do the remaining work in four bounded work packages. After each package, freeze HEAD and perform an adversarial review before starting the next.

Work package	Claude Code objective	Independent review gate
A. Package integrity	Fix src-layout packaging; build/install wheel; black-box API smoke tests; latest HEAD green on 3.11 and 3.12.	Can an external environment use the package without repo-local path tricks?
B. Research framing	Refactor README and live paper around authority-context synthesis; preserve historical corrections append-only.	Do claims now match evidence without overselling?
C. Cost-sensitive domain	Preregister realistic observation costs; execute weighted synthesis; preserve negative outcome if contracts do not differ.	Are costs defensible and registered before result?
D. Large non-planted domain + reproduction	Build/preregister 30-50-property realistic domain; run solver; prepare reproducibility packet.	Does the method generalize beyond planted and author-controlled cases?

Suggested Claude Code instruction for every work package

```
Rules: 1. Do not alter any frozen prior result. 2. Registration precedes implementation for any new
scientific result. 3. Separate bug fixes from new evidence. 4. Do not tune parameters after seeing
outcomes. 5. Preserve negative findings. 6. Run the full release-check before commit. 7. Produce a
machine-readable result artifact and a concise human summary. 8. Do not change scientific wording
unless the underlying evidence changed. 9. Stop if a result contradicts the preregistration and report
the contradiction. 10. Never use prose to compensate for a missing test or certificate.
```

8. What not to do

- **Do not write Paper 6 yet.** Consolidate this work into one stronger contribution.
- **Do not add another SARC acronym.** The research programme already has sufficient vocabulary.
- **Do not add an LLM into the compiler path.** Keep contract synthesis deterministic and machine-checkable.
- **Do not spend more research effort on the grant mechanism.** Keep it as integration engineering; capability systems already cover the primitive.
- **Do not chase a more interesting result by modifying a preregistered domain after execution.**
- **Do not add more benchmark dimensions before packaging and external reproduction are solved.**
- **Do not bury the code/cloud multiple-reduct result.** It should become a flagship example.

9. Final 9.5 checklist

Gate	Pass condition
Latest CI	Python 3.11 and 3.12 green at the same HEAD
Wheel	sdist + wheel build; fresh-venv install succeeds
API	authority_compiler imports and derives a canonical contract outside repo root
Mutation	Hard-gated and >=0.85 after all changes
Paper	Centered on authority-context synthesis, realistic multiple reducts and solver-backed contracts
README	New user sees compiler + benchmark + flagship results immediately
Realistic cost result	Minimum-cost and minimum-cardinality contract differ in at least one preregistered realistic domain, or a preregistered negative result is retained and clearly interpreted
Large realistic domain	>=30 candidate properties, non-planted, preregistered
Symbolic solution	Certified sufficient contract without exhaustive 2^n enumeration
AuthorityBench	3 existing domains remain green; larger domain can be added as fourth if appropriate
External reproduction	Independent environment reproduces at least one canonical end-to-end result
Claims	No theorem, novelty or empirical claim exceeds the evidence

9.5 decision rule

If every gate above passes, I would rate the repository approximately 9.5/10. If packaging or independent reproduction is missing, I would keep it around 9.2-9.3 even if additional scientific features are added.

10. The target end-state

The repository should emerge as a reusable governance-synthesis reference implementation with this conceptual flow:

```
GOVERNANCE INTENT | v LOSS / INVARIANTS | v EXECUTABLE REACHABILITY | v DISCERNIBILITY MODEL |
+---------------------------+ | | v v CORE ATTRIBUTES REDUCT SYNTHESIS | +-------------+--------------+
| | | v v v MIN CARDINALITY MIN COST ALTERNATIVES | | | +-------------+--------------+ | v AUTHORITY
CONTRACT | +------------------+------------------+ | | | v v v GATE SCHEMA CERTIFICATES COUNTEREXAMPLES
| v RUNTIME ENFORCEMENT
```

The final positioning should be broader than the historical paper title:

Recommended research thesis	Minimal Sufficient Governance Context: compile declared losses and reachable execution semantics into exact, cost-aware runtime authority contracts for autonomous systems.

That framing now matches what the repository is becoming: not simply a detector of relevant authorization fields, but a compiler from governance intent to sufficient runtime context, with alternative contracts, costs, counterexamples and machine-checkable evidence.

Current score: 9.2/10. **Target:** 9.5/10 after consolidation and independent evidence.